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Study of Aging and Degradation of Fluoroelastomers in HPHT Sour Conditions Under Acceleration Testing

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Abstract

Fluoroelastomers are widely used in high-pressure, high-temperature (HPHT) oil and gas downhole applications due to their resistance to aggressive chemical environments. This study investigates the aging and degradation behavior of laboratory-formulated tetrafluoroethylene-propylene (FEPM) rubbers subjected to HPHT sour (H_2S) conditions through accelerated aging tests at 200°C, 225°C, and 250°C.

Mechanical testing revealed that tensile strength and modulus at 25% and 50% strain declined significantly with aging time and temperature, indicating early onset degradation. Dynamic Mechanical Analysis (DMA) showed reductions in the glass transition temperature (T_g), suggesting crosslink density loss. Solid-state ^{13}C CP-MAS NMR confirmed rapid decline of the Triallyl Isocyanurate (TAIC) carbonyl peak near 150 ppm, supporting the observation of early-stage crosslinking breakdown. Fourier-transform infrared (FTIR) analysis further evidenced carbonyl degradation.

Combining thermal, mechanical, spectroscopic, and analytical characterization, the degradation mechanism was elucidated and the material's service life was estimated using the Arrhenius approach per ISO 23936-2 guidelines. The findings offer critical insights into FEPM behavior in sour gas environments and support the development of more robust sealing materials for extreme HPHT sour applications.

Introduction

As exploration and production shift toward deeper, harsher oil and gas reservoirs, elastomeric materials are increasingly exposed to extreme temperature, pressure, and chemically aggressive environments. Non-metallic seals, particularly fluoroelastomers, play a vital role in ensuring long-term equipment integrity under such conditions.

Fluoroelastomers are high-performance synthetic rubbers characterized by fluorinated monomer backbones, which confer excellent resistance to heat, oxidation, hydrocarbons, and corrosive agents.

Among these, fluoroelastomer types such as FKM (composed of vinylidene fluoride, hexafluoropropylene, and tetrafluoroethylene) and FEPM (a copolymer of tetrafluoroethylene and propylene, e.g., Aflas®) are commonly employed in downhole tools, including packers, O-rings, and blowout preventers (Goodson 2008; Sugama et al. 2015; Liu et al. 2020; Lee et al. 2022; Asthana et al. 2023; Ren et al. 2024).

In HPHT wells, these elastomers are continuously exposed to sour gas (H_2S), CO_2 , brine, hydrocarbons, and fluctuating thermal and mechanical loads. While fluoroelastomer (FEPM) offer favorable chemical resistance, they are still vulnerable to thermal and oxidative degradation, particularly in sour environments. Key degradation mechanisms include crosslink breakdown, chain scission, and the loss of functional groups like carbonyls, all of which impair mechanical integrity and sealing performance.

To simulate long-term field exposure, standardized accelerated aging protocols, such as ISO 23936-2 (ISO 23936-2, 2011) have been developed. These enable predictive assessment of elastomer durability under laboratory-controlled HPHT sour gas conditions. Previous studies have shown that property deterioration correlates strongly with aging time and temperature, making crosslink density and mechanical strength critical indicators of service life.

In this work, we examine the degradation behavior of a fluoroelastomer (FEPM) using a comprehensive suite of analytical techniques - including tensile testing, dynamic mechanical analysis (DMA), FTIR, TGA, and solid-state NMR. The Arrhenius model was employed to estimate service lifetime of the fluoroelastomer under downhole conditions. This study enhances our understanding of fluoroelastomer (FEPM) performance in HPHT sour environments and informs future materials design for demanding oilfield applications.

Experimental

Materials and Sample Preparation

The fluoroelastomer investigated in this study was a FEPM compound developed at CNPC-USA with a Shore A hardness of approximately 80. Standard ISO 37 Type II dumbbell-shaped specimens were die-cut from cured rubber sheets for mechanical and aging evaluations. Prior to aging, samples were vacuum dried and immersed in the test fluid at 232°C for two days to establish baseline conditions simulating initial fluid absorption without gas exposure.

Accelerated H_2S Aging Tests

Accelerated aging tests were conducted following ISO 23936-2 standards (ISO 23936-2, 2011), using high-pressure autoclaves fabricated from Hastelloy C276 to resist corrosion in sour environments. The aging fluid was a multiphase mixture consisting of:

- Gas phase (30 vol%): 10% H_2S , 5% CO_2 , and 85% CH_4
- Hydrocarbon liquid phase (60 vol%): 70% heptane, 20% cyclohexane, 10% toluene
- Aqueous phase (10 vol%): deionized water

Samples were suspended in the liquid phase, and the autoclaves were purged and pressure-tested with nitrogen before introducing the sour gas mixture via a dip tube. The gas pressure was maintained at approximately 870 psi at room temperature. After each aging cycle, heating was stopped, and the vessels were allowed to cool overnight. Gas was vented through a caustic scrubber at a controlled rate (<15 psi/min), and nitrogen purging was performed to eliminate residual sour gas.

Aging tests were conducted at three temperatures (200°C , 225°C , and 250°C) and for multiple durations (2 to 42 days), as summarized in Table 1.

Table 1—The H₂S aging experimental matrix.

Sample name	H ₂ S aging temperature (°C)	Aging duration (days)
Fluoroelastomer #3	200	7, 14, 28, 42
Fluoroelastomer #3	225	7, 14, 28, 42
Fluoroelastomer #3	250	2, 7, 14, 28

Mechanical Testing

Tensile testing was performed on an Instron 68TM-5 universal testing machine equipped with a 5 kN load cell. All tests were conducted at room temperature using a crosshead speed of 500 mm/min. Strain was measured with a contact-type external extensometer (LTE-Class B-2). For each aging condition, five specimens were tested to ensure statistical reliability.

Thermogravimetric Analysis (TGA)

TGA was conducted on a TA Instruments Q50 system with sample masses of 5–10 mg. The analysis was carried out under nitrogen with a heating rate of 10°C/min from 30°C to 700°C, followed by a gas switch to air and further heating to 900°C at 10°C/min. This two-step process allowed for differentiation of organic decomposition and residual inorganic filler content. Prior to the test, the samples were vacuum-dried at 80 °C for 4 hours and at 60 °C for 16 hours before testing to remove residual fluids.

Dynamic Mechanical Analysis (DMA)

Rectangular specimens (35 × 12 × 3.175 mm) were tested using a TA Instruments Q800 DMA in single cantilever mode. The temperature ramp was performed from −70°C to 50°C at a rate of 3°C/min. A frequency of 1 Hz and a strain amplitude of 15 μm were used within the linear viscoelastic region (LVR). Storage modulus, loss modulus, and damping factor (Tan δ) were recorded to assess thermal-mechanical behavior. Prior to testing, the samples were dried in a vacuum oven at 60 °C at 700 mmHg vacuum for 24h to remove residual fluids.

Solid-State Nuclear Magnetic Resonance (NMR)

Solid-state ¹³C NMR analysis was performed using a Varian Infinity Plus 400 MHz spectrometer under magic angle spinning (MAS) at 4 kHz. Samples (approx. 150–180 mg, cubed to ~1–2 mm) were packed in zirconia rotors with Kel-F tips and Teflon inserts. Cross-polarization (CP) was applied with contact times of 0.8 or 1.8 ms. Chemical shifts were referred to as adamantane (29.45 ppm). Data analysis was completed using Mnova 14.3.0, with zero-filling to 8k data points. Exponential multiplication of 100 Hz was consistently applied, and Whitaker baseline correction was executed for all spectra.

Fourier Transform Infrared Spectroscopy (FTIR)

Attenuated Total Reflectance Fourier Transform Infrared (ATR-FTIR) spectra were obtained using a Thermo Scientific™ Nicolet™ iS50 spectrometer. Spectra were collected over the 4000–600 cm^{−1} range with 64 scan accumulations per sample. Prior to the test, the samples were vacuum-dried at 80 °C for 4 hours and at 60 °C for 16 hours before testing to remove residual fluids. Baseline correction was applied to all spectra, and changes in the carbonyl absorption bands were monitored to evaluate chemical degradation.

Results and Discussion

Mechanical Properties

The effect of H₂S aging on the mechanical integrity of FEPM samples was evaluated via tensile testing. Figure 1 illustrates the modulus at 50% elongation over aging time at 200°C, 225°C, and 250°C. A clear

degradation trend was observed: the modulus decreased sharply in the early aging stages and continued to decline with prolonged exposure and increasing temperature.

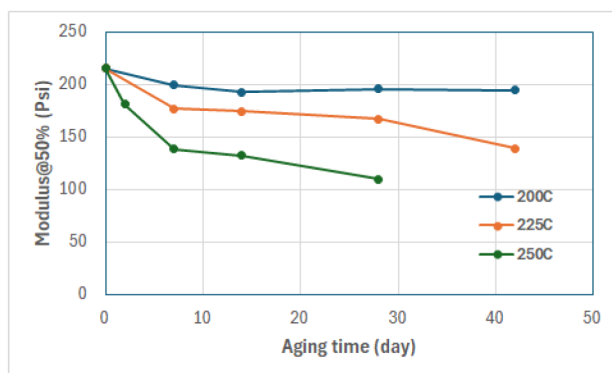


Figure 1—The modulus at 50% elongation of fluororubber #3 (80A) versus aging duration at 200 °C, 225 °C, and 250 °C.

At 250°C, samples exhibited the most rapid and severe loss in modulus, reflecting accelerated deterioration at elevated temperatures. Figure 2 presents the percent loss in modulus at 50% elongation, further highlighting the strong temperature dependence of degradation kinetics. This decline is attributed to crosslink network deterioration and chemical breakdowns induced by the reactive sour fluid.

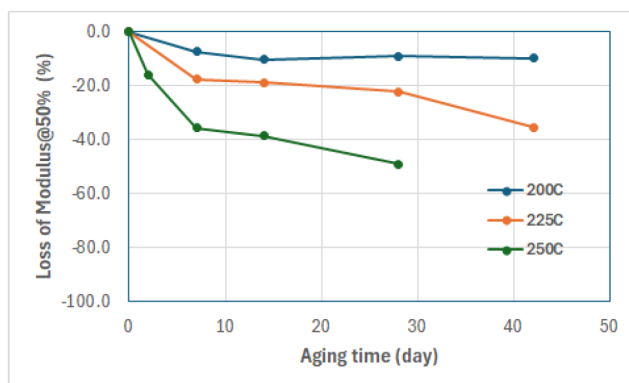


Figure 2—The loss percentage of the modulus at 50% elongation of fluororubber #3 (80A) versus aging duration at 200 °C, 225 °C, and 250°C

Thermal stability of the fluoroelastomers evaluated by thermal gravimetric analysis (TGA)

TGA is a powerful technique to evaluate the thermal stability of fluoroelastomers (Hiltz, 2014). Figure 3 compares TGA curves of unaged samples and those aged for 7 and 28 days at 225°C. Major degradation occurred between 400–520°C for all samples, with two distinct weight loss steps.

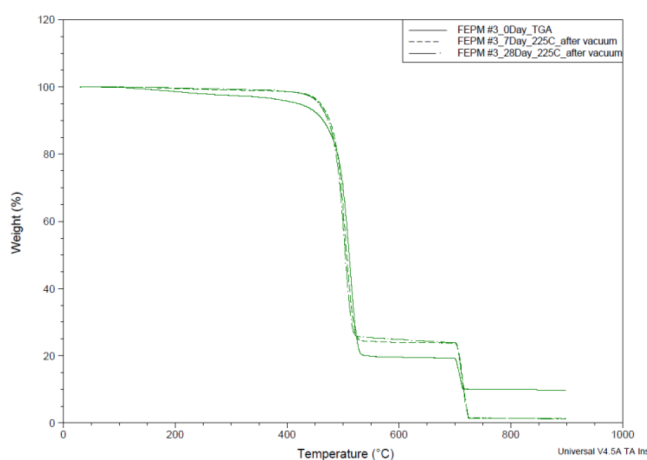


Figure 3—TGA of unaged sample and samples aged 7 and 28 days at 225 °C.

A typical TGA plot of percent weight loss versus temperature and the derivative of percent weight loss versus temperature for H₂S aged fluoroelastomer (aged at 200 °C for 14 days) are shown in Fig.4.

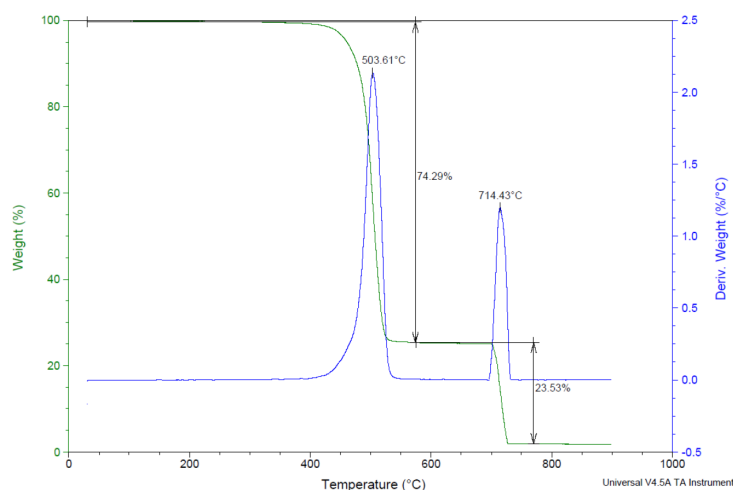


Figure 4—Plots of percent weight loss against temperature and the derivative of percent weight change against temperature for aged at 200 °C for 14 days.

The weight loss of each sample takes place in two steps. The first is at ~502 °C to ~513 °C and the second is at ~710 °C. The percent weight loss and the temperatures of the maximum rate of weight loss for the four H₂S aged fluoroelastomers at 200 °C and the weight of residue are summarized in Table 2.

Table 2—First weight loss, second weight loss, and temperatures of maximum rate of first and second weight losses of the four fluoroelastomers of H₂S aged at 200 °C for different time.

Sample name	First weight loss (%)	T _{max1} (°C) of first weight loss	Second weight loss (%)	T _{max2} (°C) of second weight loss	Residue (%)
Unaged 0 day	80.3	512.4	9.7	709.0	10.0
Aged for 7 days	75.0	502.9	23.4	713.4	1.6
Aged for 14 days	74.3	503.6	23.5	714.4	2.2
Aged for 28 days	74.4	503.7	24.0	712.6	1.6
Aged for 42 days	74.3	503.9	23.8	713.9	1.9

The unaged fluoroelastomer lost 80.3% of its weight in the first step. However, the percent weight loss in the first step was similar for H₂S aged fluoroelastomers at 200 °C for 7 days (75.0%), 14 days (74.3%), 28 days (74.4%), and 42 days (74.3%), which is about 5% to 6% higher than the unaged one. The possible reason for lower percent weight loss of the aged samples is that the organic additives were extracted out during the aging process (samples immersed in the organic solvents). The temperatures of maximum rate of weight loss, $T_{\max 1}$, for the first weight loss step were similar for H₂S aged fluoroelastomers at 200 °C for 7, 14, 28, and 42 days (502.9 °C to 503.9 °C), which was approximately 8-9 °C lower than the unaged fluoroelastomer. This suggests the fluororubber became less thermally stable after aging. The temperatures of maximum rate of weight loss for the second weight loss step, $T_{\max 2}$, were similar for unaged and aged fluoroelastomers (709 °C to 713.9 °C) (after switched gas flow to air from N₂ at 700 °C). This mainly corresponded to inorganic residue combustion.

Similar results of thermal stability (the percent weight loss in the first step and in the second step) of the fluoroelastomers aged at 225 °C and 250 °C were summarized in Table 3 and Table 4.

Table 3—First weight loss, second weight loss, and temperatures of maximum rate of first and second weight losses of the four fluoroelastomers of H₂S aged at 225 °C for different time.

Sample name	First weight loss (%)	$T_{\max 1}$ (°C) of first weight loss	Second weight loss (%)	$T_{\max 2}$ (°C) of second weight loss	Residue (%)
Unaged 0 day	80.3	512.4	9.7	709.0	10.0
Aged for 7 days	75.8	504.2	22.7	713.4	1.5
Aged for 14 days	75.3	502.8	23.4	714.4	2.3
Aged for 28 days	74.9	502.6	23.7	712.2	1.4
Aged for 42 days	75.4	502.8	23.2	714.3	1.4

Table 4—First weight loss, second weight loss, and temperatures of maximum rate of first and second weight losses of the four fluoroelastomers of H₂S aged at 250 °C for different time.

Sample name	First weight loss (%)	$T_{\max 1}$ (°C) of first weight loss	Second weight loss (%)	$T_{\max 2}$ (°C) of second weight loss	Residue (%)
Unaged 0 day	80.3	512.4	9.7	709.0	10.0
Aged for 7 days	75.5	505.0	23.2	714.0	1.5
Aged for 14 days	75.7	503.5	22.7	713.2	1.6
Aged for 28 days	73.2	503.1	24.2	715.2	2.6
Aged for 42 days	73.1	503.4	25.3	713.5	1.6

Glass transition temperature behavior by Dynamic Mechanical Analysis (DMA)

Dynamic Mechanical Analysis (DMA) is a powerful technique used to characterize elastomeric materials. The typical DMA results include plots of storage modulus, loss modulus, and tan delta versus temperature. The glass transition temperature (T_g) can be determined either from the onset of the storage modulus curve, or the peak of the loss modulus curve, or the peak of the Tan delta curve. Normally, the peak in the loss modulus curve corresponds to the temperature where the material exhibits the most significant energy dissipation due to molecular motion, often associated with the glass transition temperature (T_g). For simplicity, T_g was determined from the peak of the loss modulus versus temperature in this study.

Figure 5 presents the loss modulus curves for fluoroelastomer samples aged at 225°C for various durations (unaged, 7, 14, 28, and 42 days).

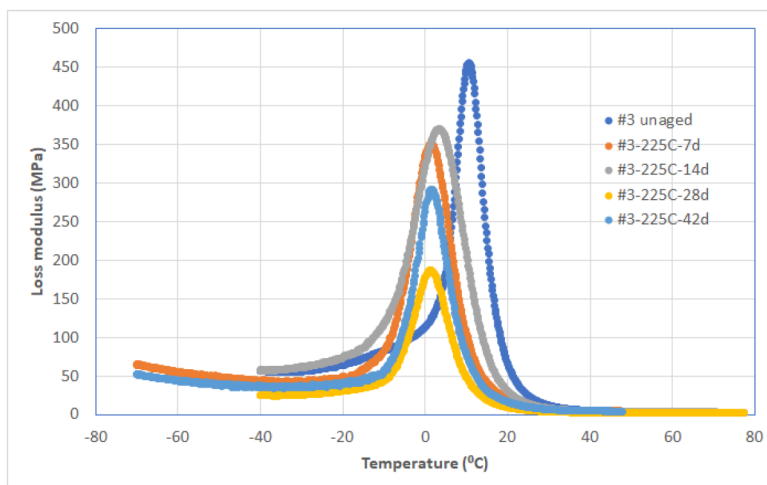


Figure 5—Comparison of Loss modulus of fluoroelastomer aged at 225 °C for different durations.

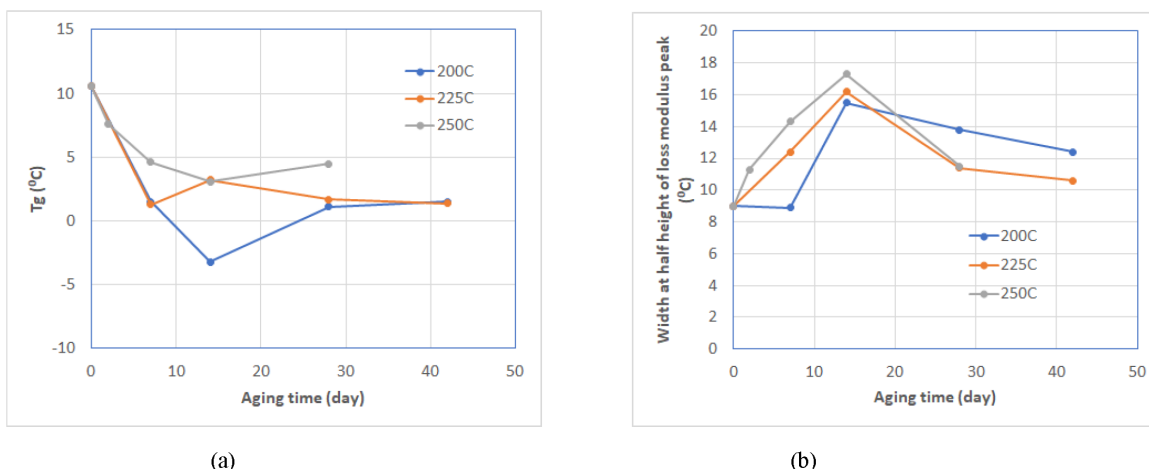


Figure 6—(a) T_g (glass transition temperature) of the fluoroelastomer aged at 200 °C, 225 °C and 250 °C for different durations. (b) The width at half height of loss modulus peak of the fluoroelastomer aged at 200 °C, 225 °C and 250 °C for different durations.

Figure 6(a) shows the T_g values derived from the loss modulus peaks of samples aged at 200°C, 225°C, and 250°C. Comparing with unaged fluoroelastomer (#3), all aged samples show decreased T_g at different aging temperatures and durations, which indicated the decrease of crosslinking density. From figure 6 (a), for T_g plot of aged at 200 °C (blue curve), the T_g decreased progressively up to 14 days, then T_g bumped up for aged 28 days and 42 days. This rebound of T_g may be attributed to extraction of the degraded small molecules portion from the sample matrix (reduced plasticization). The other possible reason is the potential re-crosslinking at high temperature and prolonged aging exposure.

In DMA, the width at half height of the loss modulus peak is a useful parameter to provide insights into the distribution of relaxation times and the homogeneity of the material's structure. For crosslinked rubbers, the width at half height of the loss modulus peak can be used to assess the uniformity of the crosslinking network. A broader peak can indicate a more heterogeneous crosslinking structure, while a narrower peak suggests a more uniform structure. The width at half height of the loss modulus peak of unaged and aged samples at 200 °C, 225 °C, 250 °C for different durations were approximately determined and plotted in Figure 6 (b). Comparing with unaged sample, the aged samples show increased the width at half height of the loss modulus peak for each aged temperature (200 °C, 225 °C, 250 °C) (Figure 6 (b)). They reach largest width at half height at aged 14 days for all three temperatures (200 °C, 225 °C, 250 °C), which indicates most heterogeneous crosslinking structure due to the degradation of crosslinking and degraded

low molecules portions in the samples. The width at half height at 28 days and 42 days became narrower than the one at 14 days, which may be attributed to the extraction of the degraded small molecules portion.

¹³C CP-MAS Solid-State NMR study

Solid-state NMR experiments are an excellent tool to nondestructively probe the bulk atomic structure of materials and give valuable information about functional groups present, structural dynamics, and changes to the electronic environment of nuclei. ¹³C CP-MAS NMR, in particular, allows the investigation of the changes on the carbon backbone of carbon-based polymers due to aging degradation.

The NMR data for aged fluoroelastomer #3 at 200 °C, 225 °C and 250 °C show very similar results. A typical ¹³C CP-MAS NMR spectra of fluoroelastomer aged at 225 °C show in Figure 7. It's clearly seen that compared with the unaged sample, the sample aged 7 days did show big loss in the carbonyl peak at ~150 ppm (from the coagent (crosslinker) Triallyl Isocyanurate (TAIC) carbonyl). This indicates that the major de-crosslinking caused by degraded TAIC happened at early stage of aging at 225 °C (7 days). Similar degradation (de-crosslinking) results at early stage were also found for sample aged at 200 °C for 7 days and sample aged at 250 °C for 2 days. In Figure 7, the aliphatic carbon region shows an increase in the CH and CH₂ peaks and little change to the CH₃. In the aliphatic region, an increase in the CH and CH₂ peaks is observed, while the CH₃ peak experiences little to no change. It is important to note that the increase in the CH and CH₂ peaks may be attributed to more efficient Cross Polarization (CP) dynamics rather than actual alterations in the polymer's structure. However, to ascertain the true source of this change, additional experiments are necessary.

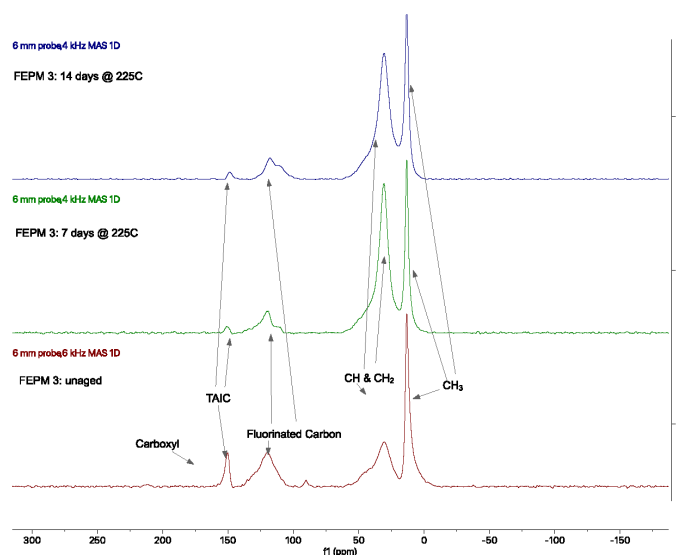


Figure 7—CP-MAS NMR spectra of unaged and aged fluoroelastomers at 225 °C (unaged, aged 7 days and 14 days).

The comparison of CP-MAS NMR spectra of aged fluoroelastomers for 28 days at 200 °C, 225 °C and 250 °C is shown in Figure 8. It's clearly seen that for the 28 days aging duration, the carbonyl peak at ~150 ppm became smaller with increase temperature from 200 °C to 225 °C and to 250 °C. This indicates that the higher aging temperature, the more degradation (de-crosslinking) for the same aging duration.

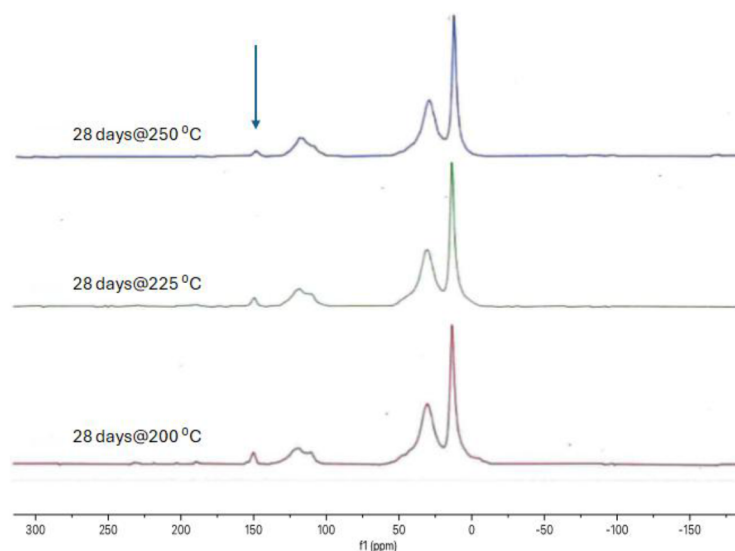


Figure 8—CP-MAS NMR spectra of aged fluoroelastomers for 28 days at 200 °C, 225 °C and 250 °C, respectively.

Characterization of the aged fluoroelastomers by Fourier transform infrared spectroscopy (FTIR)

The degradation of the aged fluoroelastomers was further investigated by integrating the FTIR. The FTIR spectra were collected with Attenuated Total Reflectance Fourier Transform Infrared (ATR-FTIR) mode from the surface of vacuumed dried samples. Figure 9 shows the overlays of FTIR spectra of unaged sample and aged samples at 250 °C for 2 days, 7 days, 14 days and 28 days. The FTIR absorbance peak around 1680 cm^{-1} is characteristic carbonyl stretching ($\text{C}=\text{O}$), where FTIR absorbance peak around 1557 cm^{-1} is asymmetric stretching of the carboxy group (COO^-). Comparing with unaged sample, both characteristic absorbance peaks (1680 cm^{-1} and 1557 cm^{-1}) decreased from aged 2 days and 7 days and reached most reduction at aged 28 days. These reduction of both characteristic absorbance peaks (1680 cm^{-1} and 1557 cm^{-1}) further confirmed the degradation of the carbonyl group (or TAIC) or de-crosslinking of the fluoroelastomer, especially at early aging durations.

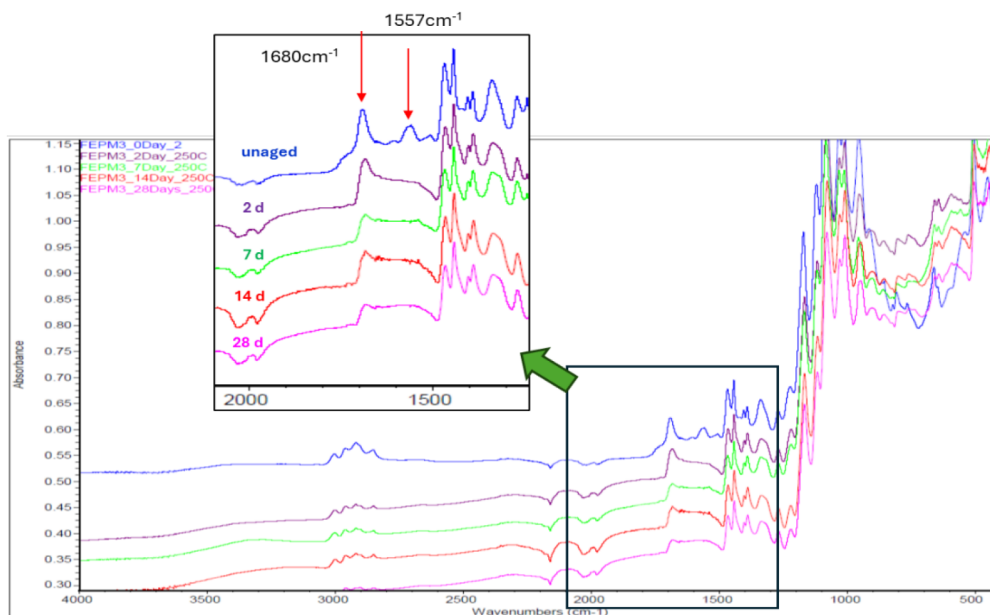


Figure 9—FTIR (ATR) spectra of unaged sample and aged samples at 250 °C for various durations (2 days, 7 days, 14 days and 28 days)

Service Life Prediction of fluoroelastomer using the Arrhenius Equation

Elastomeric materials (including fluoroelastomers) play a critical role in engineering systems due to their flexibility, sealing capability (performance), and resistance to various chemicals. They are extensively used in demanding industries such as automotive, oil and gas, and aerospace, where long-term reliability under harsh service conditions - including high temperatures, oxidative environments, and mechanical stress - is essential. However, prolonged exposure to these conditions leads to degradation of the rubber's mechanical and chemical properties through mechanisms such as thermal aging, oxidative chain scission, and crosslinking or chain cleavage. As a result, predicting the service life of elastomers has become a vital aspect of materials engineering and product reliability (Plota et al., 2020; Lee et al., 2022; Yun et al., 2023).

One of the most widely accepted approaches for predicting the service life of rubber materials involves accelerated aging tests conducted at elevated temperatures, with the results extrapolated to service conditions using the Arrhenius equation (ISO 23936-2, 2011). This method is based on the thermodynamic principle that chemical reaction rates increase exponentially with temperature. The Arrhenius equation is given by:

$$k = Ae^{\frac{-E_a}{RT}}$$

where k is the reaction rate, A is the pre-exponential factor, E_a is the activation energy, R is the universal gas constant, and T is the absolute temperature. By conducting accelerated aging tests at multiple elevated temperatures, degradation rates or property changes—such as tensile strength loss, elongation reduction, or compression set increase—are measured, and an Arrhenius plot ($\ln k$ vs. $1/T$) is used to extrapolate the degradation kinetics to typical service temperatures. (Dwivedi et al. 2014),

This approach relies on three key assumptions: (1) the degradation mechanism remains consistent across all temperatures tested; (2) the degradation reactions follow first-order or pseudo-first-order kinetics; and (3) the aging processes—such as oxidation, embrittlement, and crosslinking are thermally activated and remain consistent over the entire temperature range. Violations of these assumptions, such as changes in degradation pathways or environmental influences, can introduce significant errors in prediction. Despite these limitations, the Arrhenius method remains a practical and widely used tool for estimating the service life of elastomeric materials (ISO 23936-2, 2011).

In the present study, the modulus at 50% strain was selected as the degradation indicator for service life estimation (Figure 1). Accelerated aging tests were performed at 200°C, 225°C, and 250°C, and the modulus values over time were plotted in Figure 10. From these data, activation energies and degradation rate constants were derived, allowing service life projections at various temperatures (Table 4). The estimated service life at 140°C exceeds 25 years, whereas at 200°C it drops to less than one year (ISO 23936-2, 2011).

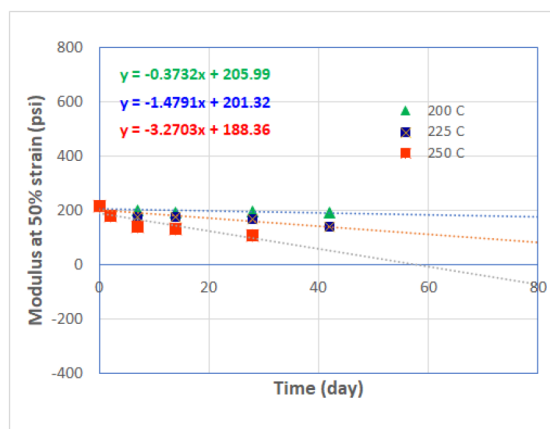


Figure 10—Experimental values of Modulus at 50% strain over time under 200 °C, 225°C, and 250°C.

Table 5—Service life estimation for the fluoroelastomer via Arrhenius equation.

Temperature (°C)	Estimated longevity (years)
140	25.2
150	12.8
160	6.8
170	3.7
180	2.0
190	1.2
200	0.7

However, the predicted service life of under one year at 200°C appears to underestimate actual performance. In real-world applications, such as downhole packer sealing element, similar fluoroelastomers have been observed to function effectively for over 20 years. This discrepancy may arise from the complex aging mechanisms in high-temperature, fluid-rich H₂S environments, which likely exceed the simplifying assumptions of the Arrhenius model. Additionally, a size effect may be at play: tensile testing was conducted on aged ~2mm thick specimens, while actual sealing components often have wall thickness exceeding 15mm, which could slow degradation due to thermal gradients and diffusion limitations.

Further work is required to bridge the gap between service life estimates derived from the Arrhenius equation and the actual performance of fluoroelastomers in downhole applications. Nonetheless, these estimates offer a foundational framework and serve as valuable guidance for the selection and specification of fluoroelastomers in sealing systems, particularly when used as comparative benchmarks. While the Arrhenius-based approach is useful, it should be applied with careful consideration of its underlying assumptions and validated against real-world service conditions to ensure reliable material performance in the field.

Conclusion

This study comprehensively evaluated the aging and degradation behavior of a FEPM fluoroelastomer developed at CNPC-USA under simulated HPHT sour conditions using a multi-technique analytical approach. Key findings include:

- Significant early-stage reductions in tensile properties and modulus at 50% strain, particularly at elevated temperatures (225–250°C), indicating rapid mechanical property degradation.
- Thermal analyses (DMA, TGA) confirmed the progressive loss of crosslink density, reduction in glass transition temperature, and thermal stability decline as aging advanced.
- Spectroscopic techniques (¹³C CP-MAS NMR and FTIR) revealed the chemical basis for degradation, including breakdown of the TAIC carbonyl group, particularly under more aggressive aging.
- The Arrhenius-based lifetime prediction method demonstrated that service life is highly temperature-dependent, with durability at 140°C estimated at over 25 years, but dropping below one year at 200°C. The estimated result for 200°C is significantly lower than the real field application, which indicated the limitation of the model.

Collectively, these results provide critical insight into the structural and property evolution of fluoroelastomer (FEPM) in sour gas environments and validate the importance of both physical and chemical assessments in service life prediction. The findings will support the development of next-generation sealing materials tailored for extreme oilfield service.

Acknowledgments

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